15. In friction experiment, if a horizontal force of 0.6 N is required to set the block of mass 0.2 kg in motion, the coefficients of static friction μ_s equal to:

a) 0.6 b) 0.2 c) 0.3 d) 0.4

16. Two identical blocks (with the same mass) are pulled along a rough surface as suggested in the figure,



Which one of the following statements is false?

(a) The coefficient of kinetic friction is the same in each case.
(b) A force of the same magnitude is needed to keep each block moving.
(c) A force of the same magnitude was required to start each block moving.
(d) The normal force exerted on the blocks by the surface is the same for both blocks.
(e) The magnitude of the force of kinetic friction is greater for the block on the right.

17. In friction experiment, if the coefficient of kinetic friction is 0.3 on a horizontal surface, the force required to keep a block of mass 0.2 kg moving with constant speed is equal:

a) 2.94 N b) 0.60 N c) 14.7 N d) 0.90 N

18. If F is the force, x the displacement, and k a spring constant, for simple harmonic motion F is given:

a) $F = kx$ b) $F = -\frac{k}{x}$ c) $F = -kx^2$ d) $F = -kx$

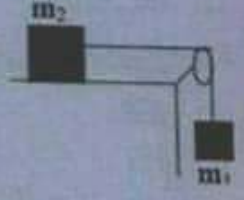
19. In elasticity experiment, if the hanging mass 80g and the elongation in the spring is 4cm, the spring constant (N/m) is:

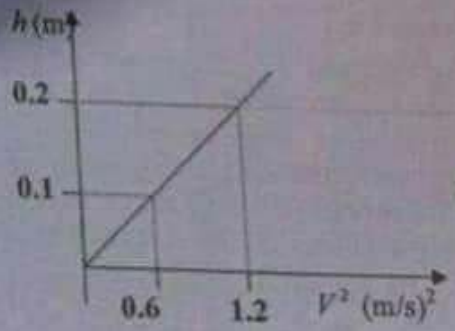
a) 5 b) 40 c) 20 d) 5

20. In elasticity experiment, if spring constant $k = 20$ N/m and the time of ten oscillations is 4.44sec, the mass of hanging weight (in gram) is:

a) 100 b) 195 c) 225 d) 75

21. In the conservation of energy experiment, a student obtained data plotted in the following figure. The masses m_1=0.15 kg and m_2=0.2 kg. The acceleration of gravity is:



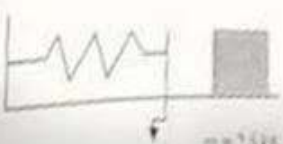


a) 7 m/s² b) 11 m/s² c) 9 m/s² d) 13m/s²

13. Suppose that the work done by an engine is given by $W = 7t^2 + 40t + 100$, where the units are SI. The power (in watt) developed by this engine at $t = 2$ s is:

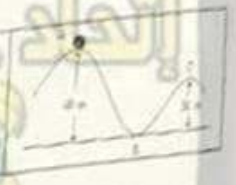
a) 68 b) 82 c) 96 d) 54

14. A mass m = 2.5 kg is sliding along a frictionless table with initial speed v as shown in the figure. It strikes a spring that has a force constant k = 1000 N/m and compresses it from its equilibrium position a distance 5 cm (x_f = -5cm). The initial speed v of the block was (in m/s):



a) 1.22 b) 1.41 c) 0.71 d) 1

15. A sphere of mass m slides on a frictionless path ABC. If the sphere starts from rest at point A, the speed of the sphere (in m/s) at point C is:



a) 24.5 b) 17.3 c) 14.1 d) 31.6

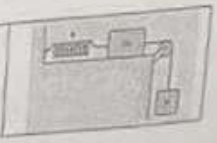
16. The potential energy function for a system is given by $U(x) = -x^3 - 2x^2 + 3x$. The magnitude of the force (in N) at x = 3 m is:

a) 92 N b) 17 N c) 36 N d) 61 N

17. A particle (m = 4 kg) moves along the x axis. Its position varies with time according to $x(t) = t^2 + 2t$, where x is in meters and t is in seconds. The kinetic energy of the particle at time t = 3 s:

a) 200 J b) 128 J c) 72 J d) 32 J

18. The blocks shown are released from rest with the spring unstretched. The pulley and the horizontal surface are frictionless. If k = 150 N/m and M = 1 kg, what is the maximum extension of the spring?



a) 13.3 cm b) 26.7 cm c) 20 cm d) 6.7 cm

Q (2): Put (T) in front of the correct phrase, and (F) In front of the wrong phrase. Then move these answers to the table. (15marks)

19) (N.m) is the unit of spring constant. (F).

20) The energy is always conserved in all kinds of collisions. (F).

21) In gravitational potential energy experiment, the change in potential energy



(ΔU) Just for (m). (F).

22) There is no relation between (m) and (τ) for pendulum at the same length. (T).

23) In an elastic collision ($(K.E)_i=(K.E)_f$) always. (T).

24) In gravitational potential energy experiment, when (m) reach the ground, the speed of (M) will increase.

(T).



25) The benefit of air track is to get frictionless surface. (T).

26) 

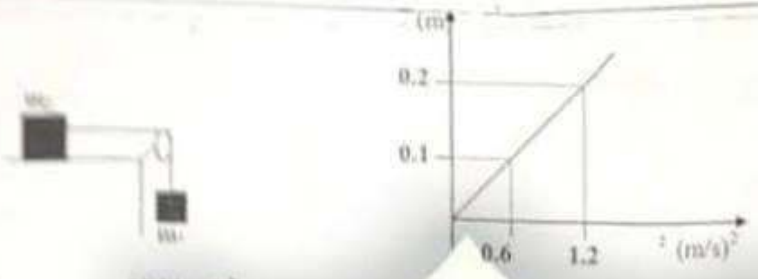
(ΔL= 4.44 cm). (F).

27) The benefit of wire Clip (crocodile clip) in uniform circular motion is to keep the radius constant. (F).

28) The number (5.000025) contains (7) significant figures. (T).

Good luck for all

13. In the conservation of energy experiment, a student obtained data plotted in the following figure. If the masses m_1 =0.15 kg and m_2 =0.2 kg, the acceleration of gravity is:



a) 7 m/s² b) 11 m/s² c) 9 m/s² d) 13m/s²

14. In the collision between two carts, all of the velocity was transferred from the first cart to the second cart. What type of collision is this?

a) Perfectly inelastic collision b) Inelastic collision c) Elastic collision d) momentum is not conserved

15. In the collision between two carts, the carts combined after the collision with one third of the velocity of the first cart. How much kinetic energy is lost if each cart has a mass m and the initial velocity of the first cart was v? (v_2 = 0)

a) $\frac{1}{3}mv^2$ b) 0 c) $\frac{8}{9}mv^2$ d) $\frac{7}{18}mv^2$

16. In uniform circular motion, if the radius of circular path is 20 cm and time needed for 50 periods was 25 sec, the Speed of the object and its centripetal acceleration (in m/s).

a) 0.63 b) 1.26 c) 2.51 d) 5.03

Q2: **Simple Harmonic Motion Experiment**

a) Calculate the spring constant from the following Data:

Load mass m (kg)	Period of oscillation T (s)	Spring constant k (N/m)
0.1	0.27	
0.2	0.40	
0.3	0.51	

b) Calculate the average value of the spring constant?

$\bar{k}$ = _____

Al- Balqa' Applied University
Faculty of Engineering Technology
Department of Applied Science

جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية
قسم العلوم التطبيقية

Physics 111
Final Exam
Time: 90 minutes (C)

الاسم بالعربية: مدرس المختبر: موعد المختبر:

A) A hanging mass M_h = **0.5kg**, and the mass of the cart M_c =1kg. The system is released from rest. The height of the hanging mass is **h=50 cm**.



a) Complete the following table:

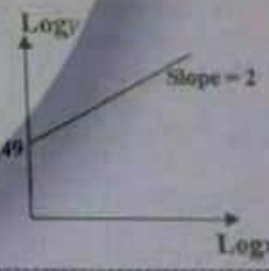
Where K is the kinetic of the system, U is the potential Energy.

U_i (J)	U_f (J)	K_i (J)	K_f (J)

The speed of the cart v=.....m/s, as the hanging mass just hits the ground.

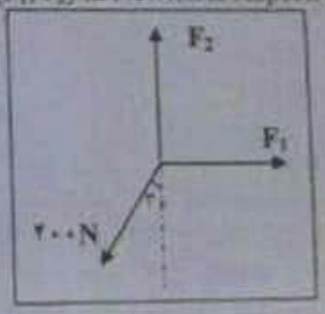
B) Circle the correct answer

1. The figure shows the relation between Log x and Log y, then:



a) $Y = 2X^3 + 2$ b) $Y = 3X^2$
c) $Y = 10X^3$ d) $Y = 3X^{10}$

2. The set of forces shown are in equilibrium, the Values of (F_1, F_2) in Newton is respectively given by:



(a) (100,173) (b) (173,100)

(c) (-100,-173) (d) (-173,-100)

3. A Vernier caliper was used to measure the length of a cube; the error in length (ΔL) is:

a) 0.1mm b) 0.05mm c) 0.5mm d) 0.005mm

4. Which of the following measurements represents the reading of the micrometer?

a) 9.25 mm b) 8.0 cm c) 1.50 cm d) 2 mm

5. A student wanted to calculate the density (ρ) of Mass (M). If $M \pm \Delta M = (10 \pm 0.01)$ gram and its volume $V \pm \Delta V = (5 \pm 0.1)\ cm^3$, the absolute error of the density ($\Delta\rho$) in g/cm^3 is:

a) 0.08 b) 0.04 c) 0.06 d) 0.02

6. An object is moving with a constant speed in a circular path of radius r. If its frequency of revolution is (2n), express its centripetal acceleration in terms of n and r.

a) $4\pi^2n^2r$ b) $16\pi^2n^2r$ c) $8\pi^2n^2r$ d) $2\pi^2n^2r$

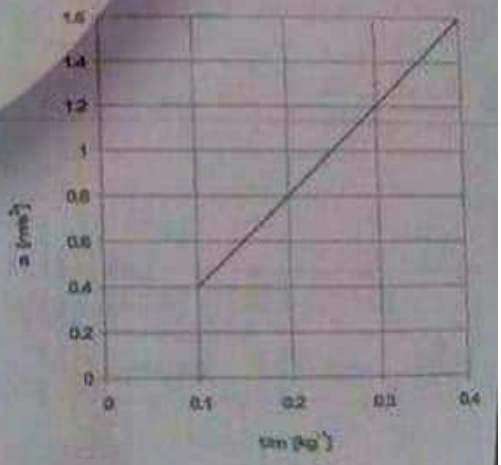
7. In the circular motion experiment, a student studied the relation between the hanging mass (M) and the square of the frequency (f^2) for rotational body (m) by making the radius (R=0.6 m). A plotted data is shown in the figure bellow.



The mass (m) of the rotational body is:

a) 6.34 gram b) 12.7 gram c) 8.94 gram d) 4.22 gram

In the Newton's second law experiment, the relation between the acceleration (a) and the total mass (m) with a constant force (F) is shown in Figure.



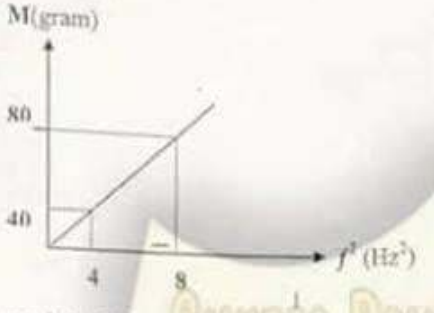
8. The applied force on the system (in N) is:

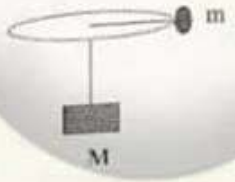
a) 4 b) 12
c) 16 d) 8

9. The hanging mass m_h (in gram) is:

a) 800 b) 120
c) 400 d) 160

8. In the circular motion experiment, a student studied the relation between the hanging mass (M) and the square of the frequency (f^2) for rotational body (m) using the radius (R=0.6 m). A plotted data is shown in the figure bellow.

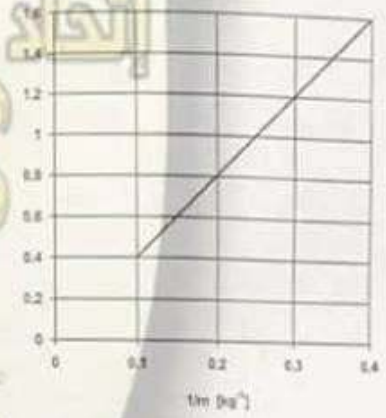




The mass (m) of the rotational body is:

a) 6.34 gram b) 12.7 gram c) 8.44 gram d) 4.22 gram

In the Newton's second law experiment, the relation between the acceleration (a) and the total mass (m) with a constant force (F) is shown in Figure.



9. The applied force on the system (in N) is:

a) 4 b) 12
c) 16 d) 8

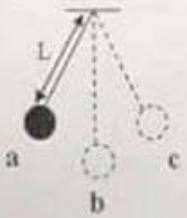
10. The hanging mass m_s (in gram) is:

a) 800 b) 120
c) 400 d) 160

In the figure shown, the bob takes 0.749 seconds to move from point a to b. If the length of the string is 2 m, then

11. The acceleration due to gravity, g, (in m/s^2) is:

a) 9.2 b) 8.8
c) 10 d) 9.6



12. If we doubled the mass of bob, then the period (T) will be:

a) 2T b) T c) T/2 d) $(T)^{1/2}$

Student Polytechnic Union
SPU

Al- Balqa' Applied University
Faculty of Engineering Technology
Department of Physics and Basic Sciences

جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية
قسم العلوم الفيزياء والعلوم الأساسية

Physics 111
Final Exam
Time: 90 minutes (B)

الاسم بالعربية: مدرس المختبر: موعد المختبر :

Q1) Circle the correct answer **(32 degrees)**

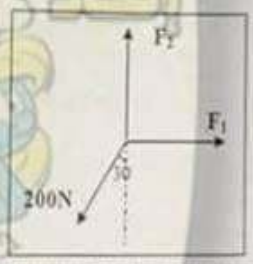
1. Two forces F_1 , F_2 are equal to each other and to their resultant, the angle between the two forces is:

a) 120^0 b) 45^0 c) 180^0 d) 90^0

2. The set of forces shown in the following figure are in equilibrium, the magnitudes of F_1 and F_2 (in N) are respectively given by:

(a) (100,173) (b) (173,100)

(c) (-100,-173) (d) (-173,-100)



3. A Vernier caliber was used to measure the length of a cube, the error in length (ΔL) is given by:

a) 0.1mm b) 0.05mm c) 0.5mm d) 0.005mm

4. Which of the following measurements represents the reading of the micrometer?

a) 7.28 mm b) 8.0 cm c) 1.59 cm d) 2 mm

5. In spring constant experiment, if the hangings mass 80g and the elongation in the spring is 2cm, the spring constant (N/m) is:

a) 5 b) 40 c) 20 d) 5

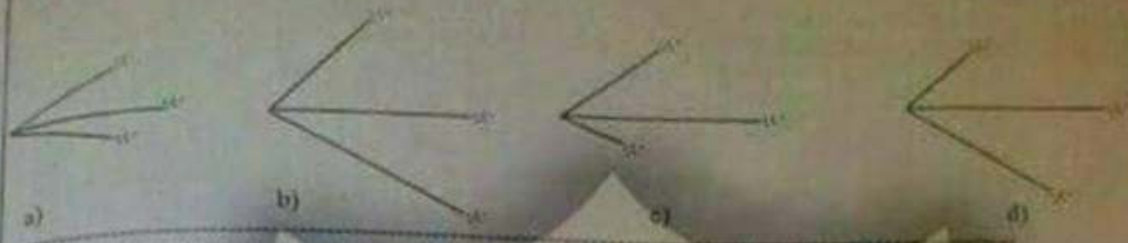
6. A simple pendulum has a simple harmonic motion if:

a) The string is very long.
b) The angle gives $\sin\theta \approx \theta$.
c) The mass to length ratio is small.
d) The mass to length ratio is large.

7. In friction experiment, if the coefficient of kinetic friction is $\mu_k = 0.2$, the force required to keep a block of mass 0.3 kg moving horizontally with constant speed is equal to:

a) 2.94 N b) 0.60 N c) 14.7 N d) 0.90 N

10. In an elastic collision of two steel balls of equal masses, which of the following represents the correct experimental situation?

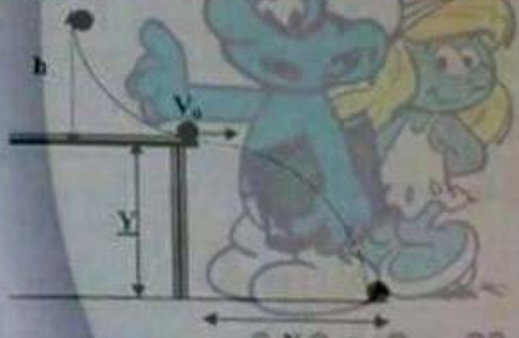


a) b) c) d)

11. One of the following statements is **true**

a) In the elastic collisions, the linear momentum is conserved, but the kinetic energy is not.
b) In inelastic collisions, the linear momentum and the kinetic energy are not conserved.
c) The linear momentum is conserved for both elastic and inelastic collisions.
d) The kinetic energy is conserved for both elastic and inelastic collisions.

12. In the collisions in two dimensions experiment, a student obtained the results as shown in the figure for the mass M_1=10g when dropped from a height h to calculate its velocity before collisions with another mass, for Y= 80cm, and X= 40 cm.



The velocity (Vo) is:
a) 2.0 m/s b) 1.0 m/s c) 4.0 m/s d) 1.4 m/s

The height h equals _____

13. The motion of a simple pendulum is approximately simple harmonic if and only if:
a) The string is very long.
b) The angle gives $\sin\theta \approx \theta$.
c) The mass to length ratio is small.
d) The mass to length ratio is large.

14. In the simple pendulum experiment, the following figures show two pendulums. If the periodic time is T_1 for the Pendulum (1), and the periodic time is T_2 for the pendulum (2). Then the relation between T_1 and T_2 is



a) $T_2 = 2T_1$ b) $T_2 = 1.73T_1$ c) $T_2 = 2.24T_1$ d) $T_2 = 2.45T_1$

5) In an inelastic one dimension collision (v_{1i}=5 m/s),(v_{1f}=2 m/s),(v_{2i}= 0)and (M_2= 6 g).then (M1) is:

A) 2 g	B) 4 g	C) 6 g	D 8 g

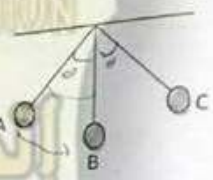
6) The magnitude of (R) is:



A) 3 N	B) 5 N	C) 6 N	D) 9.4 N

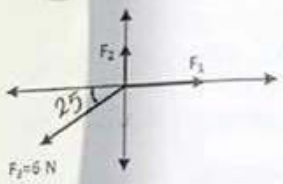
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7) As($\Theta_{AB}=\Theta_{BC}$),and (t) from (a to b)=0.06,then (τ)is :



A) 0.06 s	B) 0.12 s	C) 0.24 s	D) 0.3 s

8) For an equilibrium system the magnitude of (F_2) is:

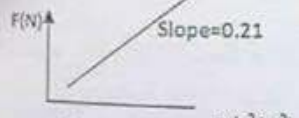


A) 2.5 N	B) 3N	C) 5.2 N	D) 5.4 N

9) In Hooks law experiment (m) oscillates under the effect of spring, which has the spring constant (k), if the spring changed to another one with spring constant (4k), the period of one oscillation (τ) will be:

A) 4 τ	B) τ/2	C) 2 τ	D) none of all

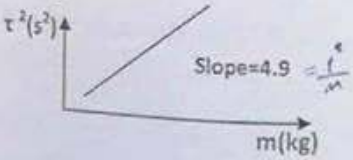
10) In Newton's second law experiment, if (M_{Total}=300g), the distance Which (M_w) moved in all trials, is:



$\frac{F t^2}{2}$

A) 70 cm	B) 80 cm	C) 50 cm	D) 60 cm

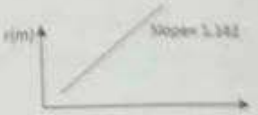
11) In Hooks law experiment, the spring constant (K) is:-



A) 9 N/m	B) 8 N/m	C) 6 N/m	D) 7 N/m

$\tau^2 = \frac{4\pi^2 m}{K}$

12) In uniform circular motion, if (m=10g),(M=45g),then the accelera on of gravity (g) is:



A) 9 m/s^2	B) 10 m/s^2	C) 8 m/s^2	D) 11 m/s^2

13) In one dimension an elastic collision, ($M1=2M_2$), ($V_{1i}=3m/s$), ($v_{2i}=5m/s$), and the linear momentum after collision ($p_f=8$ kg.m/s). Then (M_2) is:



A) $M_2=8$ kg	B) $M_2=16$ kg	C) $M_2=30$ kg	D) $M_2=28$ kg

14) In newton's second law experiment,(Mw=160g) ,(m=40g) then the accelera on is:



A) 1 m/s^2	B) 1.5 m/s^2	C) 2 m/s^2	D) 2.5 m/s^2

15) In this vector addi on, if the scale used to draw it (2mm=2N), Then the magnitude of resultant is:



A) 0	B)10 N	C)15 N	D) 20 N

16) In uniform circular motion experiment, if the radius (r=50cm), (M=40g),(m=12g),the velocity of (m) is:

A) 4.1 m/s	B) 4.5 m/s	C) 3.2 m/s	D) 3.7 m/s

17) A spring with the length of (10 cm), we hanged amass (60g) to the end of it, which made the spring to stretch and the length became (10.3cm),the spring constant is:

A) 200 N/m	B) 400 N/m	C) 150 N/m	D) 250 N/m

18) If the time of (20) oscillations in hook law's experiment for amass (50 g) is (6 s). The spring constant (K) is:

A) 12 N/m	B) 22 N/m	C) 32 N/m	D) 49.3 N/m